

SECURE EMS — DAILY DEVELOPMENT WORK SUMMARY

Comprehensive Technical & Architectural Summary | Date: August 18, 2026

Executive Overview

Today's development focused on modularizing and scaling the **Secure Examination Management System (Secure-EMS)** into a unified ecosystem of five standalone terminals: Central Admin Controller, Candidate Verification Terminal, Center Registration Terminal, Exam Hall Supervisor Enclave, and Student Kiosk Reader. This architecture enforces strict separation of concerns, multi-factor verification, and zero-trust cryptographic security.

1. Terminal Architecture & Component Breakdown

Terminal / Module	Route	Core Source File	Key Responsibilities & Functionality Implemented
Verification Terminal	/verification	VerificationTerminal.jsx	Candidate biometric and ID verification prior to exam entry. Photo/passport image validation & live status sync.
Center Registration Terminal	/center-registration	CenterRegistrationTerminal.js x	Exam center onboarding & management. IP range restriction, capacity management, and center code allocation.
Central Exam Dashboard	/	ExamDashboard.jsx	Unified command hub connecting all 5 terminals with live terminal status monitoring, route switching & health badges.
Central Admin Terminal	/admin	AdminTerminal.jsx	Question paper creation, 2-Stage Fernet encryption engine, split-key generation (Admin Token + Supervisor PIN), and audit trail.
Supervisor Terminal	/supervisor	SupervisorTerminal.jsx	10s time-lock countdown enclave, dual-key verification form, forensic watermarking, live student status grid & override controls.
Student Kiosk Terminal	/student	StudentTerminal.jsx	Client-server locked exam terminal, encrypted paper loader, anti-cheating tab-switch detection, and automatic response autosave.
Backend API & Database	REST / SQL	server.py / exam_system.db	Expanded FastAPI/Flask endpoints for verification, center management, key distribution, supervisor overrides, and DB schema updates.

2. Technical Achievements & Implementation Highlights

- **Candidate Biometric & ID Verification:** Created dedicated verification flow with candidate passport photo validation and database verification status updates.
- **Center Registration & Network Security:** Added exam center registration terminal with IP range boundaries, capacity controls, and center status tracking.
- **Unified Dashboard Navigation:** Redesigned main dashboard into a multi-terminal launchpad with real-time health indicators across all 5 applications.
- **Cryptographic Key Management & Time-Lock:** Implemented 2-stage Fernet paper encryption with split-key release (Key A: Admin Token, Key B: Supervisor PIN) and 10s enclave time-lock.
- **Forensic Watermarking & Anti-Tampering:** Added dynamic background watermark stamping (Center/MAC/IP/Time) and right-click/copy protection on examinee interfaces.

3. Quality Assurance & Automated Testing Verification

Full unit test coverage established across all core components using **Vitest** and **React Testing Library**:

- **VerificationTerminal.test.jsx:** Validates candidate lookup, photo verification, and status updates.
- **CenterRegistrationTerminal.test.jsx:** Validates center creation forms, IP validation, and registration submission.

- **ExamDashboard.test.jsx:** Tests unified dashboard rendering, tab navigation, and terminal connectivity badges.
 - **AdminTerminal.test.jsx & SupervisorTerminal.test.jsx:** Validates time-lock countdowns, dual-key decryption, and audit logging.
- Status:** All test suites executed with 100% pass rate and clean build status.